

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY CHANGA

Third Semester of B. Tech. (Backlog) Examination (CL)
May 2011

CL 204 Engineering Geology

Date: 20.05.2011, Friday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Draw neat figures wherever required. Also quote suitable examples.

SECTION - I

- Q - 1 What is an earthquake? Explain the terms with neat sketch: Epicenter, Epicentral distance, Seismic zone and Focal depth. [07]
- Q - 2 (a) Define: Petrology and Mineralogy. Write about the scope of Geology in Water Resource Engineering. [05]
- (b) Write about erosion and weathering process. Explain the frost and thermal action of weathering on the rock mass. [05]
- (c) Define and explain briefly the following terms: [04]
- (i) Cleavage in minerals
- (ii) Jolly's spring balance method for specific gravity

OR

- Q - 2 (a) How Sedimentary and Igneous rocks are recognized in the field? Enlist these rocks. Enlist mechanical structures developed in Sedimentary rocks. [05]
- (b) Write short notes on: [05]
- (i) Rock forming minerals and Ore minerals
- (ii) Oxidation and Reduction process on rocks
- (c) Explain the hardness of minerals and Moh's scale along with hardness of some accessories used in laboratory. [04]
- Q - 3 (a) Give the Hatch scheme of classification of Igneous rocks. How colour scheme and textures in Igneous rocks are helpful to identify the rocks? [05]
- (b) (i) Write a note on scope of geology in construction field. [03]
- (ii) Write about texture in igneous rocks. [02]
- (c) (i) Explain briefly transportation of sediments by wind? [02]
- (ii) Write briefly the magnitude of an earthquake. [02]

OR

- Q - 3 (a) Give the generalized classification of Metamorphic rocks. Also write about zones and grade in Metamorphic rock. [05]
- (b) Explain geological works of underground water. Enlist various features developed due to these works. [05]
- (c) Explain briefly the methods of transportation of sediments by water. [04]

SECTION – II

- Q - 4 (a) What is mass movement? How water has influence on mass movement? [05]
- (b) Draw a neat sketch of an ideal fault showing various parts. [02]
- Q - 5 (a) What is fold and folding in rock? Describe causative factors of fold. [05]
- (b) Write a note on engineering consideration of an unconformity and joint. [05]
- (c) Write a short note on "Occurrence of joints in various rocks". [04]

OR

- Q - 5 (a) What is fault? Write a note on causative factors for the fault. [05]
- (b) Differentiate Anticline and Syncline fold. Also write about Symmetrical and Asymmetrical type of fold. [05]
- (c) What is the influence of different water levels for site selection of reservoir. [04]
- Q - 6 (a) Write a note on earthquake forces affecting the dam structure in water filled and empty condition of reservoir. [05]
- (b) What are the geological investigations carried out for selecting tunnel alignment? Add your note with tunnel in soft rock formation. [05]
- (c) What is siltation problem in reservoir structures? State the precautionary measures and remedial measures to check the siltation problems. [04]

OR

- Q - 6 (a) Define following terms: [05]
Aquifer, Aquitard, Aquiclude, Artesian aquifer and Perched aquifer.
- (b) Write a note on suitability of site with Anticline and Syncline type of fold for the dams. Also suggest your opinion. [05]
- (c) Write a note on suitability of site with a fault and shear zones in areas for the tunnel. [04]
